David Phan

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Summary of Qualifications

Programming languages/software: Java, Python, Bash/Command Line scripting, C, C++, Jupyter Notebook, Scala, VS Code, HTML, CSS, Javascript/Typescript, React, Bootstrap 4, Dagger DI, Mockito
Version control/Workflow optimization software: Git, Github, Git Actions, Docker Compose
Database and Cloud service software: AWS, Azure, MongoDB, PostgreSQL
Leadership/Project Management: Engineering Ambassador at UW, ECE Tutor at UW, Officer for Music for Charity at UW, School Music Director, Communications Lead at Washington Aerospace Scholars (WAS)

Education

University of Washington, Seattle Sept. 2020 - June 2024
Bachelor of Science in Electrical and Computer Engineering (ECE) GPA: 3.87, Cum Laude

Experiences and Activities

SDE I - Amazon Sept. 2024 - Present

- Operated Amazon's core highly-available distributed media ingestion service, managing operational tasks such as oncall rotations and peak event preparation.
- Modernized Amazon's content ingestion by adding support for media types such as HEIF, reducing monthly incompatibility ingestion related errors by 32%.
- Enhanced team operations effectiveness by leading monthly bug bash sessions, onboarding new team members to our team oncall rotations, and migrating our team to new task management tooling.
- Migrated legacy bare-metal deployment infrastructure to CDK based AWS cloud infrastructure.

Software Development Engineer Intern - Amazon Jun. 2023 - Sept. 2023

- Spearheaded internal full-stack application project to aid ML research teams in processing datasets.
- Designed and implemented both a React/Typescript front-end and a REST API using AWS API Gateway, Lambda, s3, DynamoDB, and Amazon internal tools.
- Deployed applications through CI/CD frameworks using Git, internal pipeline tools, and AWS Pipelines.
- Collaborated with a team of developers to generate and meet design goals using the Agile Methodology/Framework.

Software Development Intern - Makeability Lab, Project Sidewalk Mar. 2022 – Jun. 2024

- Maintain and update the Project Sidewalk API and website, which tracks and updates geographical information on pedestrian accessibility issues in current city infrastructure.
- Collaborate with a team of software developers to respond to GitHub tickets to quickly debug issues and implement new features in the code base or frontend UI/UX.
- Utilize technologies like Scala, PostgreSQL, and other frontend languages/frameworks.

Undergraduate Researcher - Theberge (BCME) Lab Jul. 2022 – Jun. 2024

- Automated data-analytics and image processing tasks by coding python-based desktop applications.
- Designed and prototyped microfluidic and biomedical devices for research and medical purposes using CAD software, 3D printing and Milling devices, and microcontroller circuits.
- Built and optimized website features to represent lab research and expand study group participation.

Software Engineering Intern - nFlux.ai Sept. 2021 – Dec. 2021

- Analyzed and processed video files for ML training and worked with senior engineers on DevOps tasks to develop CI/CD pipelines using Docker Compose and Git Actions to enhance project workflow.
- Designed and developed python code that automated and organized database information retrieval for SQL

and noSQL databases.

- Developed functionality testing procedures to analyze UI/UX components of company products.